

KHALED YOUNES

Department of Mechanical Engineering $\diamond$ Stanford University

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EDUCATION

Stanford University, Stanford, CA

Sept. 2021 – Aug. 2026

Ph.D. in Mechanical Engineering

Thesis: Characterizing the Picosecond Dynamics of Liquid Water Subject to Hard X-ray Irradiation

Committee: Mike Dunne, Matthias Ihme, David Reis

University of Waterloo, Waterloo, ON

Sept. 2019 – Aug. 2021

M.A Sc. in Mechanical Engineering

Thesis: Velocity Scaling of High-Speed Turbulent Boundary Layer Flows with Wall Heat Transfer

Degree Honors: Finalist for the Alumni Gold Medal & Governor General's Gold Medals

University of Waterloo, Waterloo, ON

Sept. 2012 – June 2017

B.A Sc. in Mechanical Engineering

Degree Honors: With Distinction, Dean's Honors List

PUBLICATIONS

[[Google Scholar](#)]

18. **K. Younes**, H. Li, A. Majumdar, Y. Sun, T. Yoon, P. Ho, S. Wada, S. Song, T. Osaka, I. Inoue, D. Zhu, M. Dunne, D. Reis, L. Young, M. Ihme, "Acoustic fingerprints of radiation spurs in liquid water radiolysis," (Under review in Nature Physics)
17. M. Asplund, H. Liu, T. Yoon, Y. Liu, R. Wallick, S. Bhattacharyya, X. Cheng, S. Eisenberg, R. England, M. Graßl, ..., **K. Younes**, B. Fingerhut, M. Ihme, T. Elsaesser, M.-F. Lin, B. Schwartz, J. Rouxel, L. Young, "Direct measurement of the structure of the hydrated electron," (Under review in Nature Communications)
16. A. Majumdar, H. Li, Y. Sun, S. Song, G. Vignat, P. Muhunthan, T. Sato, **K. Younes**, A. Halavanau, A. Lutman, D. Sokaras, D. Zhu, M. Ihme, "Emergence of ultrafast collective dynamics from nanostructural heterogeneities in supercritical H₂O and D₂O," (Under review in Nature Communications)
15. T. Yoon, H. Li, Y. Sun, A. Majumdar, P. Muhunthan, G. Vignat, S. Song, T. Sato, S. She, J. Fan, Y. Li, **K. Younes**, A. Halavanau, A. Lutman, A. Maurya, D. Zhu, M. Ihme, "Revealing Ultrafast Multiscale Dynamics of Liquid CO₂ Radiolysis with Time-Resolved X-ray Pump/X-ray Probe Scattering using Split-and-Delay Optics," *Struct. Dyn.* (Accepted)
14. J. Fan, T. Yoon, G. Vignat, H. Li, **K. Younes**, A. Majumdar, P. Muhunthan, D. Sokaras, T. Weiss, I. Rajkovic, M. Ihme, "Supercritical Ethanol-CO₂ Mixtures Exhibit Microscopic Immiscibility: A Combined Study Using X-ray Scattering and Molecular Dynamics Simulations," *J. Phys. Chem. Lett.*, vol. 16, no. 27, 2025. doi:10.1021/acs.jpcclett.5c01293
13. P. Muhunthan, A. Majumdar, **K. Younes**, G. Vignat, H. Li, I. Rajkovic, T. Weiss, D. Sokaras, M. Ihme, "A self-consistent analysis of cluster morphology in supercritical carbon dioxide from Small Angle X-ray Scattering," *Chemical Physics Letters*, vol. 876, no. 142190, 2025. doi:10.1016/j.cplett.2025.142190

12. P. Muhunthan, H. Li, G. Vignat, E. R. Toro, **K. Younes**, Y. Sun, D. Sokaras, T. Weiss, I. Rajkovic, T. Osaka, I. Inoue, S. Song, T. Sato, D. Zhu, J. L. Fulton, M. Ihme, “A versatile pressure-cell design for studying ultrafast molecular-dynamics in supercritical fluids using coherent multi-pulse x-ray scattering,” *Rev. Sci. Instrum.*, vol. 95, no. 013901, 2024. doi:10.1063/5.0158497
11. **K. Younes**, M. Poli, P. Muhunthan, S. Ermon, M. Ihme, “Autonomous screening of complex phase spaces using Bayesian optimization for SAXS measurements,” *Nuclear Inst. and Methods in Physics Research, A*, vol. 1057, no. 168719, 2023. doi:10.1016/j.nima.2023.168719
10. T. Zirwes, G. Vignat, E. R. Toro, E. Boigné, **K. Younes**, D. Trimis, M. Ihme, “Improving volume-averaged simulations of matrix-stabilized combustion through direct X-ray μ CT characterization: Application to NH₃/H₂-air combustion,” *Combustion and Flame*, vol. 257, part 2, no. 113020, 2023. doi:10.1016/j.combustflame.2023.113020
9. G. Vignat, T. Zirwes, E. R. Toro, **K. Younes**, E. Boigné, P. Muhunthan, L. Simitz, D. Trimis, M. Ihme, “Experimental and numerical investigation of flame stabilization and pollutant formation in matrix stabilized ammonia-hydrogen combustion,” *Combustion and Flame*, vol. 250, no. 112642, 2023. doi:10.1016/j.combustflame.2023.112642
8. **K. Younes** and J.-P. Hickey, “Mean velocity scaling of high-speed turbulent boundary layer flows under nonadiabatic wall conditions,” *AIAA Journal*, 2022. doi:10.2514/1.J062547
7. **K. Younes**, B. Gibeau, S. Ghaemi, J.-P. Hickey, “A Fuzzy Cluster Method for Turbulent/Non-Turbulent Interface Detection,” *Experiments in Fluids*, vol. 62, no. 73, 2021. doi:10.1007/s00348-021-03169-9.
6. **K. Younes**, A. Grenke, J.-P. Hickey, M. Gagnon, B. Elzein, “Enhanced Delayed Detached Eddy Simulations of Shock-Vector Control,” *23rd AIAA International Space Planes and Hypersonic Systems and Technologies Conference*, AIAA 2020-2411. doi:10.2514/6.2020-2411.
5. J.-P. Hickey, **K. Younes**, M. Yao, D. Fan, J. Mouallem, “Targeted turbulent structure control in wall-bounded flows via localized heating,” *Physics of Fluids*, vol. 32, no. 035104, 2020. doi:10.1063/1.5144387. **(Featured Article; Highlighted in Scilight)**
4. **K. Younes** and J.-P. Hickey, “Fluidic Thrust Shock-Vectoring Control: A Sensitivity Analysis,” *AIAA Journal*, vol. 58, no. 4, 2020. doi:10.2514/1.J058922.
3. **K. Younes** and J.-P. Hickey, “Effects of shear layer growth on the indirect noise in compound nozzles,” *Journal of Sound and Vibration*, vol. 468, no. 115090, 2020. doi:10.1016/j.jsv.2019.115090.
2. **K. Younes** and J.-P. Hickey, “Indirect noise prediction in compound, multi-stream nozzle flows,” *Journal of Sound and Vibration*, vol. 442, pp. 609–623, 2019. doi:10.1016/j.jsv.2018.10.061.
1. J.-P. Hickey and **K. Younes**, “Path to turbulence in a transitional asymmetric planar wake,” *Physics of Fluids*, vol. 31, no. 104107, 2019. doi:10.1063/1.5118891.

RESEARCH EXPERIENCE

Graduate Research Assistant | *Stanford University*

Sept. 2021 – Aug. 2026

- Conducted hard X-ray pump hard X-ray probe scattering experiments on liquid water radiolysis at the SLAC National Laboratory and the SPring-8 Angstrom Compact LAsEr facility (SACLA).
- Post-processed large data sets (>15 TB) to extract the ultrafast ionization dynamics and quantify their impact on the formation of radiation track structures.

- Performed large-scale atomistic simulations using LAMMPS, paired with a Monte-Carlo framework, to model the multi-step ionization cascade following core ionization in liquid water.
- Developed a reduced-order analytical model to validate the numerical simulations and provide a comprehensive understanding of the radiation effects.
- Summarized and presented the findings in technical reports, international conferences, and departmental presentations, effectively communicating the concepts to specialized and broad audiences.
- Implemented Bayesian optimization to achieve a 2x increase in state sampling of complex phase spaces for supercritical fluid applications.

Research Assistant | *University of Waterloo*

July 2017 – Aug. 2021

- Devised a wall model based on energy conservation for velocity scaling in hypersonic flows with heat transfer.
- Developed low-order analytical models in Python to compute the noise generated in gas turbine nozzles and study the fluidic thrust shock-vectoring response of rocket engines.
- Conducted global variance-based sensitivity analyses to explore the full state space, identify main flow inputs, and reveal intra-variable interactions.
- Performed high-fidelity detached eddy simulations in SU² to quantify the shock-vectoring performance of hybrid rocket engines under various injection flow rates.

AWARDS & SCHOLARSHIPS

• Canada Impact+ Research Training Award (CIRTA Postdoctoral)	2026
• Early Career Award (ULITIMA Conference)	2023
• Postgraduate Scholarship: Doctoral (NSERC PGS D)	2021
• Canada Graduate Scholarships: Master's Program (NSERC CGS M)	2020
• Ontario Graduate Scholarship	2020
• President's Graduate Scholarship	2020
• Sandford Fleming Foundation Teaching Assistantship Excellence Award	2020
• Graduate Research Studentship	2020
• University of Waterloo Graduate Scholarship	2020
• Engineering Excellence Master's Fellowship	2019
• Engineering Dean's Entrance Award	2019
• People's Choice Award	2017
• John Deere Limited Scholarship	2017
• First In Class Engineering Scholarship	2016
• University of Waterloo President's Scholarship	2013
• Engineering International Student Scholarship	2013

CONFERENCE PRESENTATIONS

10. **K. Younes**, M. Poli, P. Muhunthan, S. Ermon, M. Ihme, "Rapid, Online Screening of Complex Phase Spaces Using Bayesian Optimization with Application to SAXS Measurements," *15th International Conference on Synchrotron Radiation Instrumentation*, Hamburg, Germany, Aug. 2024. **(Invited)**

9. **K. Younes**, H. Li, Y. Sun, S. Song, T. Osaka, I. Inoue, M.-F. Lin, P. Ho, D. Zhu, M. Ihme, “Observing the Ultrafast Structural Response of Liquid Water Subject to Strong XFEL Radiation,” *15th International Conference on Synchrotron Radiation Instrumentation*, Hamburg, Germany, Aug. 2024.
8. **K. Younes**, H. Li, Y. Sun, S. Song, T. Osaka, I. Inoue, D. Zhu, M. Ihme, “Ultrafast Structural Reorganization of Supercritical Water Molecules Under XFEL Radiation,” *APS Division of Atomic, Molecular and Optical Physics*, Fort Worth, TX, Jun. 2024.
7. **K. Younes**, M. Poli, P. Muhunthan, S. Ermon, M. Ihme, “Rapid, online screening of complex phase spaces using Bayesian Optimization for SAXS measurements,” *Ultrafast Imaging and Tracking Instrumentation, Methods and Applications Conference*, Menlo Park, CA, Mar. 2023. **(Student Award)**
6. **K. Younes** and J.-P. Hickey, “Mean-velocity scaling of compressible turbulent boundary layer flows under non-adiabatic wall conditions,” *APS Division of Fluid Dynamics Meeting Abstracts*, Nov. 2020.
5. J.-P. Hickey, H. Daryan, and **K. Younes**, “Dynamics of compound, compressible flow contractions,” *APS Division of Fluid Dynamics Meeting Abstracts*, Nov. 2020.
4. M. Yao, **K. Younes**, D. Fan, J. Mouallem, and J.-P. Hickey, “Targeted modal turbulent flow control via localized heating,” *APS Division of Fluid Dynamics Meeting Abstracts*, Seattle, WA, Nov. 2019.
3. **K. Younes** and J.-P. Hickey, “Effects of shear layer growth on the indirect noise in compound nozzles,” *Thousand Islands Fluid Dynamics Meeting*, Gananoque, ON, Apr. 2019.
2. **K. Younes**, M. Yao, J. Wang, and J.-P. Hickey, “Towards predictive simulations for rocket propulsion,” *Ontario Aerospace Council*, Toronto, ON, Mar. 2019.
1. **K. Younes** and J.-P. Hickey, “Indirect noise in compound-compressible nozzle flows,” *Thousand Islands Fluid Dynamics Meeting*, Gananoque, ON, Apr. 2018.

TEACHING EXPERIENCE

Graduate Teaching Assistant | *Stanford Mechanical Engineering* Jan. 2024 – Apr. 2026

- Delivered lectures and bi-quarterly review sessions to 60+ students on advanced thermodynamic concepts, as part of ME 30: Engineering Thermodynamics and ME 357: Gas Turbine Analysis.
- Prepared and graded examinations, regularly monitoring student progress and ensuring fair assessment.
- Held weekly help sessions to provide one-on-one support on course material.

Graduate Teaching Assistant | *Waterloo Faculty of Engineering* Jan. 2020 – Aug. 2021

- Responsible for running tutorial sessions and marking exams for 85+ third-year mechanical engineering students, as part of ME 353: Heat Transfer and ME 351: Fluids Mechanics I.

In-House Residence Tutor | *Claudette Millar Hall Residence* Sept. 2017 – Apr. 2018

- Tutored 10 first-year engineering students Calculus on a weekly basis.

General Teaching Assistant | *Waterloo Engineering Endowment Foundation* Jan. 2014 – Apr. 2014

- Mentored and advised 450+ first-year students, emphasizing time management and perseverance.
- Conducted weekly and monthly help sessions in Calculus and Physics for 150 students.

MENTORSHIP

- Liu Deng (Rotating Ph.D. student, now at FX Lab) Sep. 2025 – Dec. 2025
- Kshitij Ghormode (Master's student, now at NVIDIA) Sep. 2023 – Apr. 2024
- Kai Rayle (Undergraduate student, now at Amidon Heavy Industries) June 2023 – Aug. 2023

ACADEMIC SERVICE

- Reviewer for Progress in Energy and Combustion Science, Journal of Fluid Mechanics, Physics of Fluids, Journal of Aerospace Engineering, AIAA Journal, the Aeronautical Journal, IEEE Transactions on Aerospace & Electronic Systems, Shock Waves

REFERENCES

- **Professor David A. Reis**
Professor of Photon Science and of Applied Physics
Stanford University
dreis@stanford.edu
- **Professor Jean-Pierre Hickey**
Associate Professor of Mechanical and Mechatronics Engineering
University of Waterloo
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